Cabify
legitimate or artificial traffic

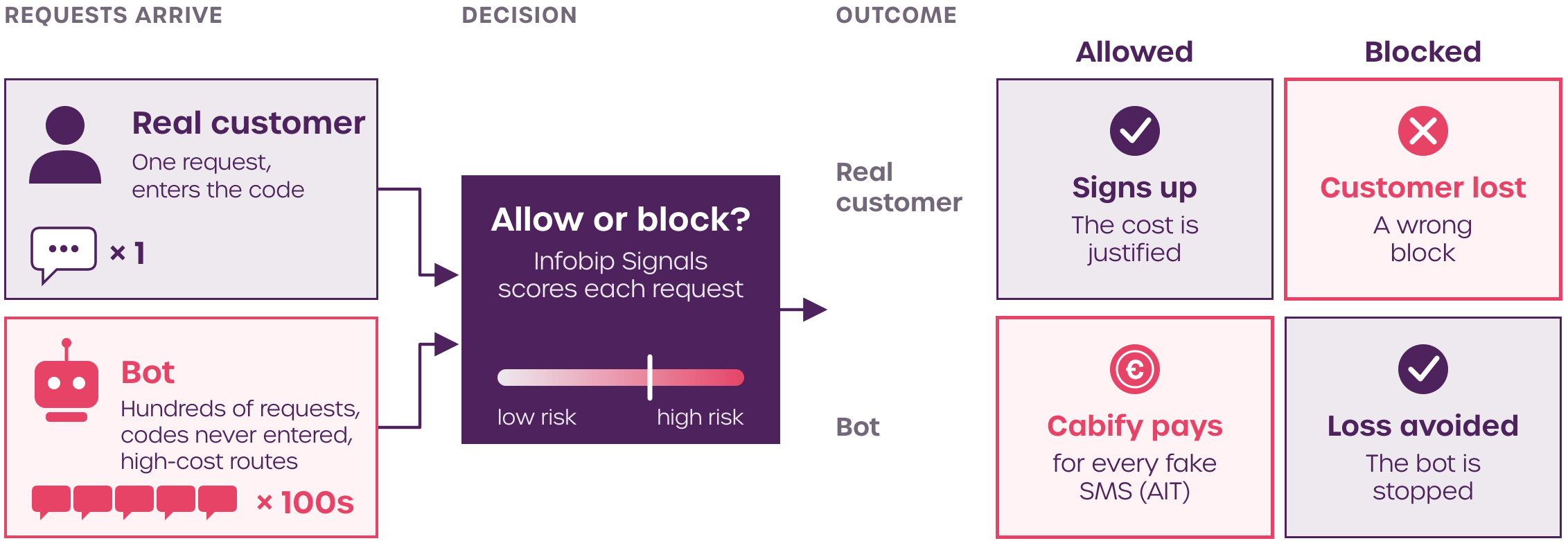
Binary classification and a blocking threshold, explained step by step

InnoIT · ML use case

Innoit

Cabify: the problem

Every verification SMS costs money, whether a person or a bot asked for it



Both mistakes cost money. The blocking threshold decides which one you accept more often.

Source: Infobip customer case with Cabify testimony

Binary classification

Cabify (Spain)

Operational service via Infobip Signals

Concept

Assign one of two labels, often by applying a threshold to a risk score.

Advantage

Automates high-volume decisions and reduces manual intervention.

Real use case

Infobip Signals distinguishes legitimate SMS from bot traffic and blocks fraudulent requests for Cabify.

Limitation

False positives can block real customers. Attack patterns change over time.

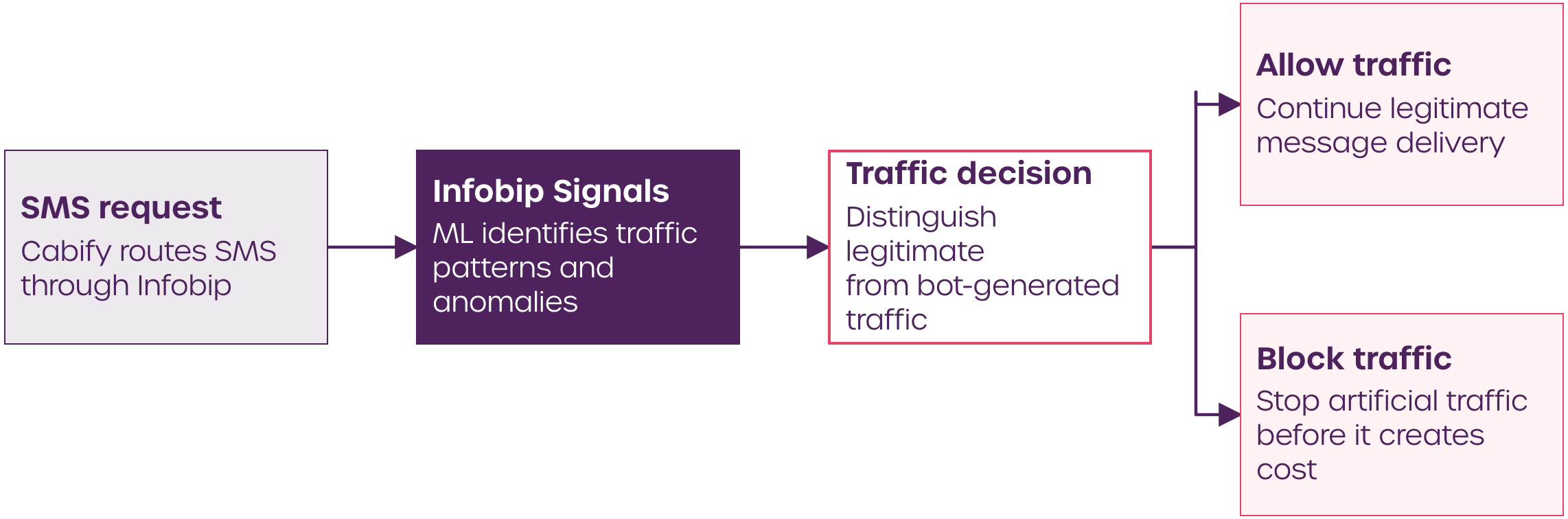
Infobip reports 15.7% less AIT-related financial loss in one quarter. Model architecture is undisclosed.

A two-way prediction becomes useful when its errors have clear business costs.

Source: Infobip customer case with Cabify testimony

Cabify: legitimate or artificial traffic

The ML service makes a concrete allow/block decision on SMS traffic



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The ML service makes a concrete allow/block decision on SMS traffic

1 Request | 2 Risk score | 3 Threshold | 4 Cost | 5 Explainability

The request becomes a feature vector

Verification request

€1.50 per unblocked bot message · fictional

Requests/min	6	Codes entered	45%
High-cost route	No	Unfamiliar device	Yes

↓

x = [1.95, 0.45, 0, 1]

ln(1 + requests/min), share of codes entered, high-cost route, unfamiliar device

The signals are illustrative. Infobip does not publish the features its ML service uses.

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Technical details

A verification SMS to check

A sign-up asks for a verification SMS. Change its signals to follow this one request through every step.

Requests per minute 6

Codes entered 45%

High-cost route ☐

Unfamiliar device ☑

Computed demo · synthetic requests and prices

← / → steps · teaching baseline · no Cabify or Infobip data or proprietary model

Source: Infobip customer case with Cabify testimony

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Learning first, scoring second

TRAINING · computed once

3,000 labelled requests
Signals X + bot or genuine y

→ **Logistic regression**
Learn weights β

→ **Maximum likelihood**
min log-loss

SCORING · updates with your inputs

This request's signals
6/min · 45% codes entered

→ **Fixed fitted model**
$p = 1 / (1 + e^{-z})$

→ **Risk of bot traffic**
68%

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Technical details

Logistic regression turns signals into a score

Learn from past labelled requests, then turn this request's signals into a probability of artificial traffic.

What does it learn?

Entering more codes lowers risk. Each doubling of the request rate adds +1.14 log-odds, holding the other signals fixed.

A fitted association in synthetic data.

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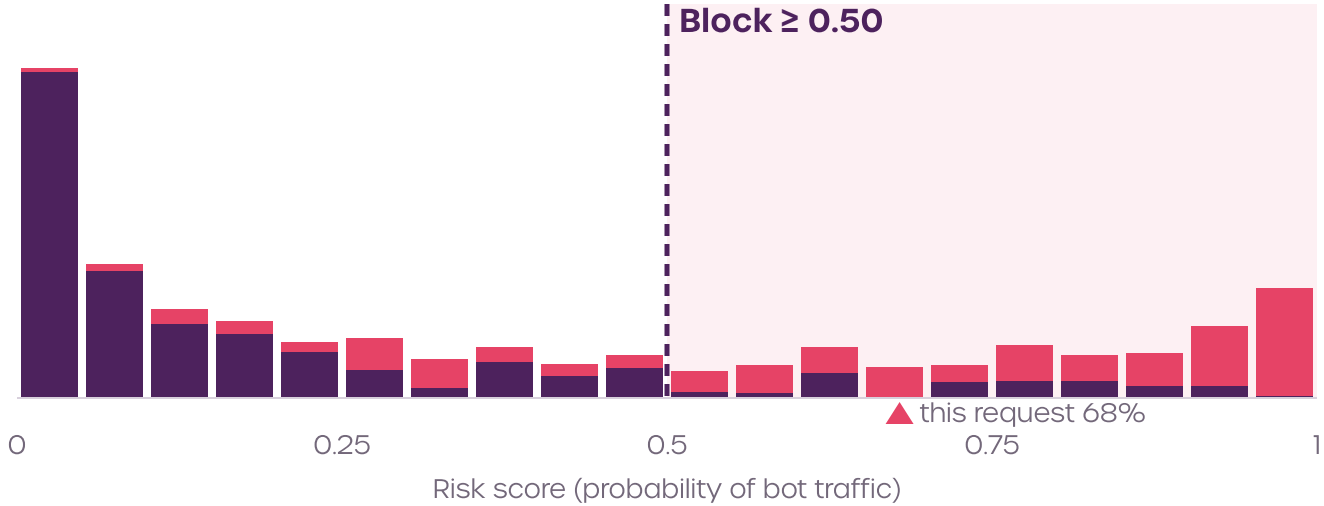
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Where the line falls on held-out traffic



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Technical details

A threshold turns the score into a decision

Choose the score at which a request is blocked. Every threshold lets some bots through and stops some genuine users.

Block at 0.50

This request: Block

Score 68% against a threshold of 50%.

Genuine users blocked: 30 of 815. Bots allowed: 51 of 185.

Computed demo · synthetic requests and prices

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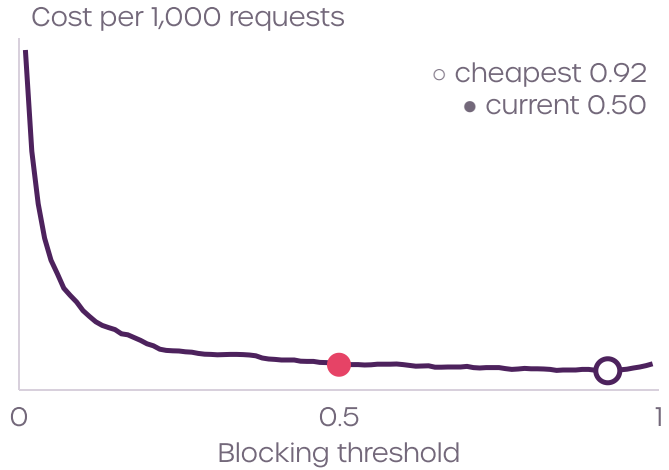
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1 Request | 2 Risk score | 3 Threshold | **4 Cost** | 5 Explainability

Counting the mistakes at this threshold

	Blocked	Allowed
Actually a bot	**134** bots stopped	**51** bots missed · €1.50 each
Actually genuine	**30** real users blocked · €5 each	**785** real users allowed



30 × €5 + 51 × €1.50, scaled to 1,000 requests **= €227**

Fictional prices, chosen to illustrate the trade-off. They are not Cabify or Infobip figures.

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Technical details

Each kind of error has a price

Put a price on each mistake. The best threshold depends on what each kind of error costs.

Block at 0.50

Cost of blocking a real user €5

Precision 82% · recall 72%

Of the requests blocked, 82% are bots. Of all bots, 72% are stopped.

Use cheapest threshold (0.92)

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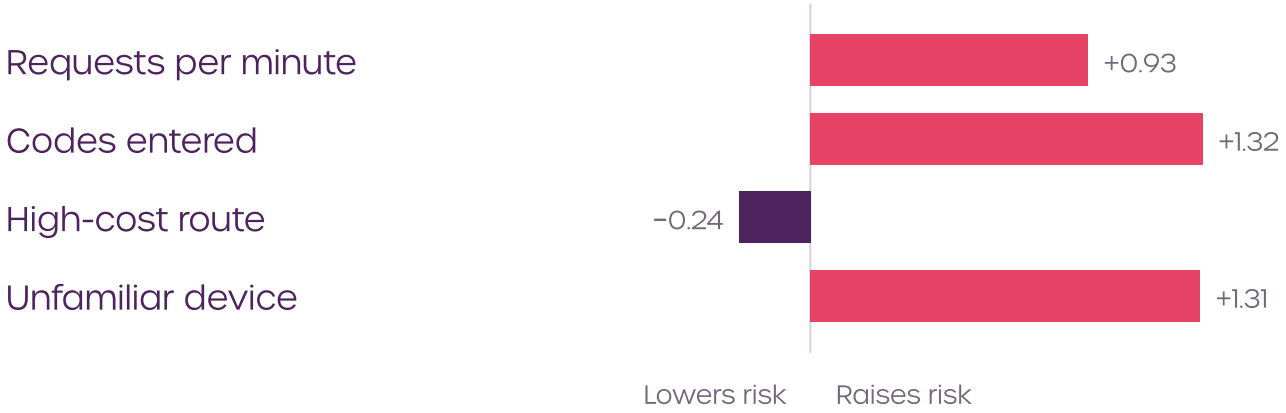
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An exact log-odds breakdown of this score



Average request −2.57 + signals **= +0.75 log-odds → 68% risk**

Exact arithmetic of this model, in log-odds. The average request is the mean of the training data.

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Technical details

Why did this request score this way?

Split the risk score into the signals that pushed it up or down for this request.

This request

Codes entered moves its risk the most: +1.32 log-odds against the average request.

Decision at 0.50

Block. The score is 68%.

Bars explain this calculation. They do not prove intent, and a high score is not proof of fraud.

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Discussion

Which decision in your business is still made by a person for every case?

What would a wrong block cost you?

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